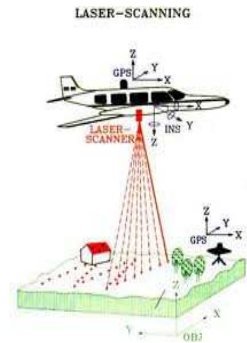


AltaLIS LiDAR15 DEM

The AltaLIS LiDAR15 DEM is a high accuracy and high resolution DEM that has been collected by LiDAR technology and processed into 15 meter post spacing, XYZ ASCII coordinates.

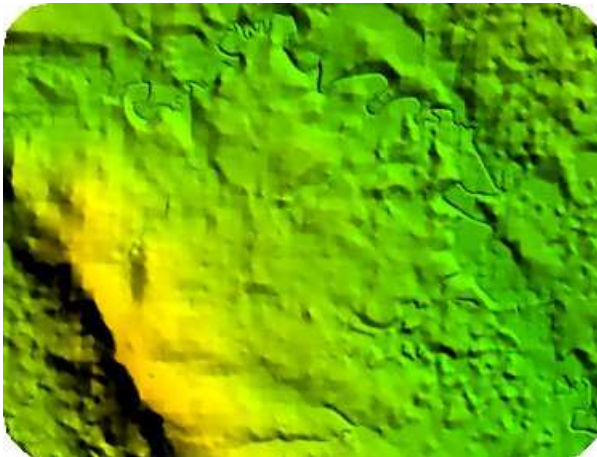
DEM collected with LiDAR (Light Detection and Ranging) technology offers a very high vertical accuracy, which enhances the quality of relief, hydrological features, sub-catchment boundaries (basins and watersheds) and drainage networks.

Bare Earth LiDAR DEM includes the returns within the point cloud from the ground, and returns from vegetation and man-made structures have been removed. Laser beams from an aircraft platform scan the ground perpendicular to the line of flight. Laser returns, or reflections from the ground and foliage are processed to determine the ground elevations. As LiDAR penetrates tree cover, it provides the most accurate ground definition possible.

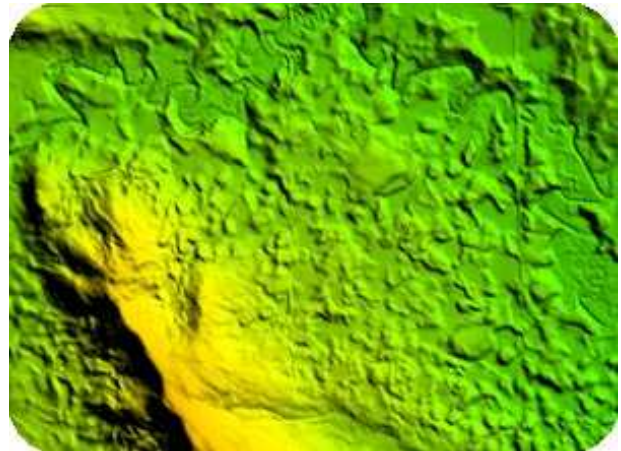


LiDAR15 DEM features **30cm** vertical, **50cm** horizontal accuracy, and is available in two formats: **ESRI Arc Grid ASCII** and **LiDAR15 ASCII**. The AltaLIS DEM product has an accuracy of **5m** and consists of grid points, break lines, and spot heights that have been compiled using 1:60 000 aerial photography.

Compare the detail:



AltaLIS DEM



LiDAR15 DEM

LiDAR15 DEM METADATA

15 meter post spacing, Bare Earth

Ellipsoid: Nad83

Projection: UTM (6TM)

Vertical Datum: CGVD28 (HT2)

Accuracy: 30cm vertical (Z) & 50cm horizontal (X/Y)

ESRI ARC GRID ASCII format:

ARC GRID ASCII refers to a specific ESRI interchange format developed for ARC Map raster data in ASCII format. The format consists of a header that specifies the geographic domain and resolution, followed by the actual grid cell values.

Example:

```
ncols      1194
nrows      981
xllcorner  654792.500
yllcorner  5735492.500
cellsize   15
nodata_value -9999
1193.06 1193.836 1194.682 1193.477 1192.764 1192.479 1192.249 1190.09 1190.109 1190.564
1190.654
```

Records 1-6 Geographic Header: Coordinates may be in decimal or integer format.

ncols: refers to the number of columns in the grid

nrows: refers to the number of rows in the grid

xllcorner: refers to the western edge of the grid

yllcorner: refers to the southern edge of the grid

cellsize: refers to the resolution of the grid

nodata value: refers to the value that represents missing data. This is optional and your parser should not assume it will be present. Note: The ESRI default is -9999.

Record 7-End of File: Data values of the individual cells that represent the elevation of an area.

$val(nox, noy) (f)$ = individual grid values, column varying fastest in integer format. Grid values are stored as floating point values.

xllcorner and *yllcorner* are given as the lower left edges of the grid, NOT the centers of the edge cells.

LiDAR15 ASCII format:

LiDAR15 ASCII DEM refers to a custom 15 meter DEM in a straight ASCII file developed in a generic format. The ASCII file displays coordinates in an XYZ space delimited format, and does not have any header information.

Example:

```
654800 5750200 1193.060
654800 5750185 1193.733
654800 5750170 1194.460
654800 5750155 1194.991
654800 5750140 1195.748
654800 5750125 1196.456
654800 5750110 1196.105
```

Records: The records are listed in an XYZ space delimited sequence. X and Y coordinates are listed as integers, and the Z coordinates are listed as decimal values.